

# ImPRESS — Time-to-Event Proportional-Hazards Assessment

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## 1 Introduction

This document provides a pre-unblinding assessment of the **proportional-hazards (PH) assumption** for the Cox models reported in the time-to-event section of the main mock-up report (`make_rdtte.R`), in **54 subjects** (cohort 2). It addresses consistency-review point **C3** (SAP states the PH assumption is assessed by standard diagnostics; the main report shows hazard ratios without the diagnostics).

For each endpoint the Cox model is `coxph(Surv(aval, 1 - cnsr) ~ trtcd)`, where `trtcd` is the long-term planned dose (0/25/50/100 mg) — the same specification used for the reported hazard ratios. The PH assumption is assessed with:

1. **Scaled Schoenfeld residual test** (`survival::cox.zph`) — a global test plus a per-term test; small p-values indicate a time-varying (non-proportional) effect.
2. **Schoenfeld residual plots** — scaled residuals against transformed time; a non-zero slope signals departure from PH.
3. **Complementary log-log survival plots** —  $\log(-\log S(t))$  against  $\log(\text{time})$  by dose; approximately parallel curves are consistent with PH.

All analyses use sham-randomised data and are descriptive.

**Interpretation guide.** Treat the global Schoenfeld p-value as the headline:  $p > 0.10$  — no evidence against PH; 0.05–0.10 — borderline, inspect plots;  $p < 0.05$  — evidence of non-proportionality, consider a time-varying coefficient, stratification, or reporting a time-averaged hazard ratio with this caveat.

## 2 Per-endpoint diagnostics

### 2.1 Overall survival (OS)

34 events in 54 subjects.

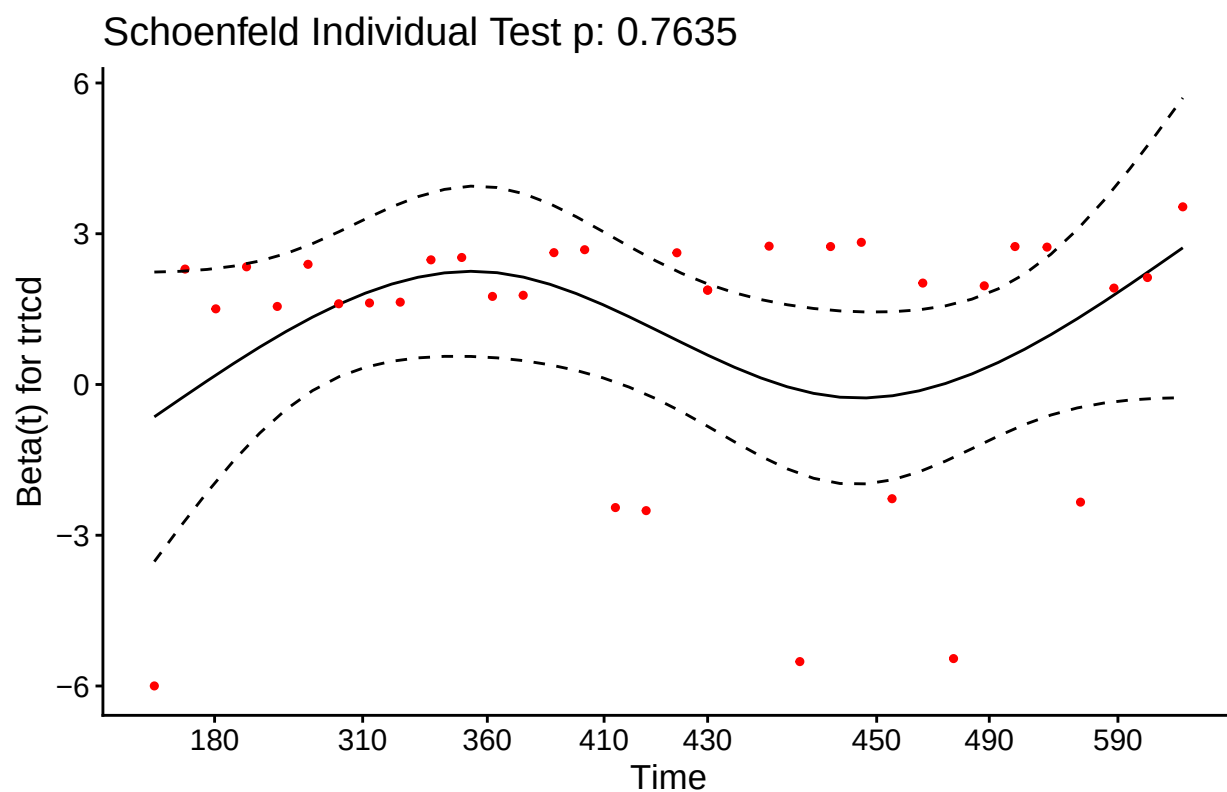
Table 1: Scaled Schoenfeld test — Overall survival

|        | Term             | Chi-square | df | p-value |
|--------|------------------|------------|----|---------|
| trtcd  | trtcd            | 1.156      | 3  | 0.7635  |
| GLOBAL | GLOBAL (overall) | 1.156      | 3  | 0.7635  |

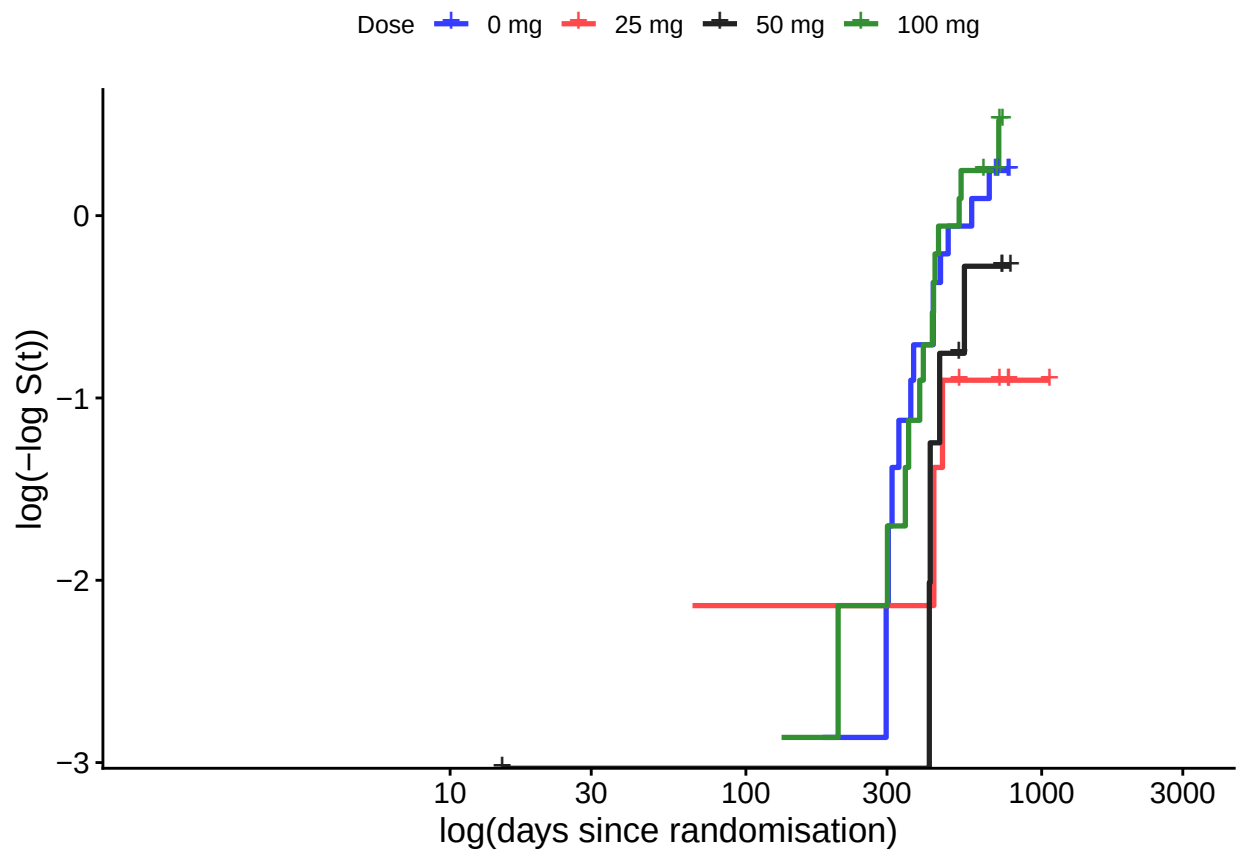
**Global Schoenfeld p-value:** 0.764 — No evidence against PH.

#### 2.1.1 Scaled Schoenfeld residuals

Global Schoenfeld Test p: 0.7635



### 2.1.2 Complementary log-log survival



## 2.2 Overall survival at 12 months (OS12M)

12 events in 54 subjects.

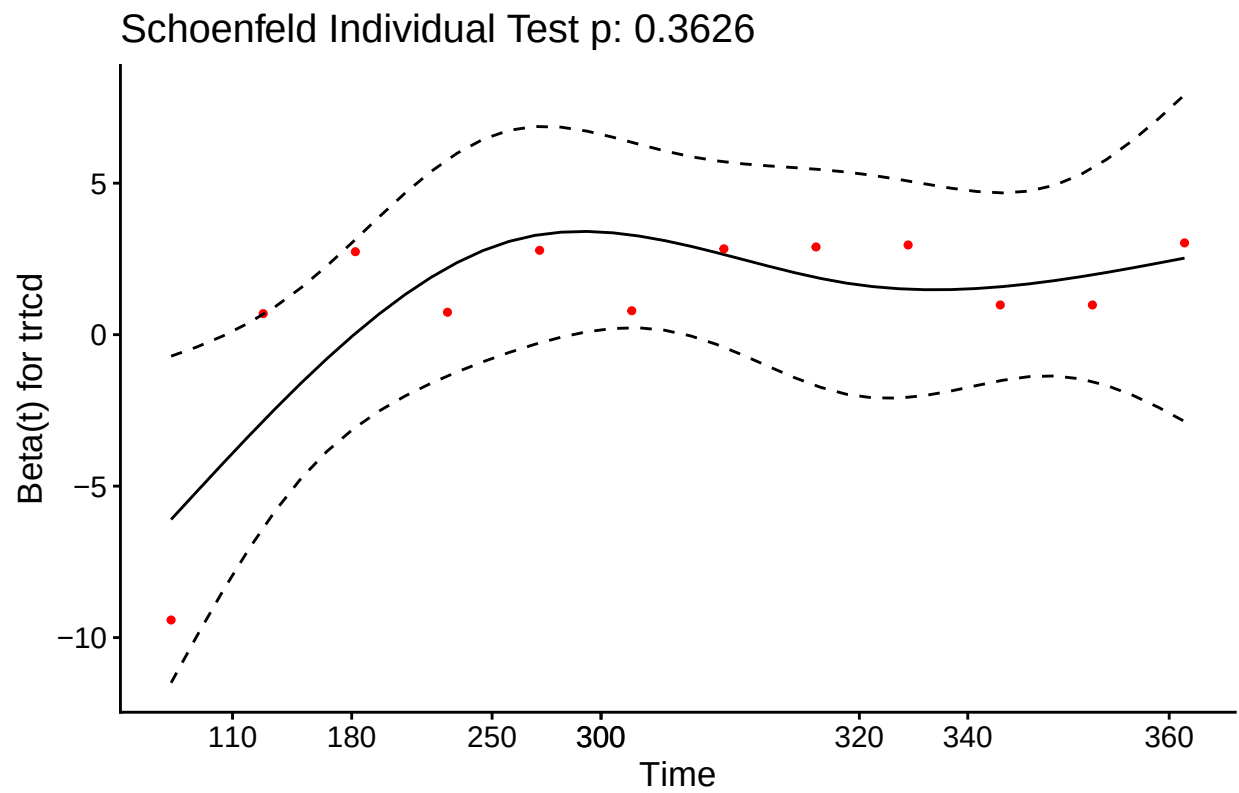
Table 2: Scaled Schoenfeld test — Overall survival at 12 months

|        | Term             | Chi-square | df | p-value |
|--------|------------------|------------|----|---------|
| trtcd  | trtcd            | 3.195      | 3  | 0.3626  |
| GLOBAL | GLOBAL (overall) | 3.195      | 3  | 0.3626  |

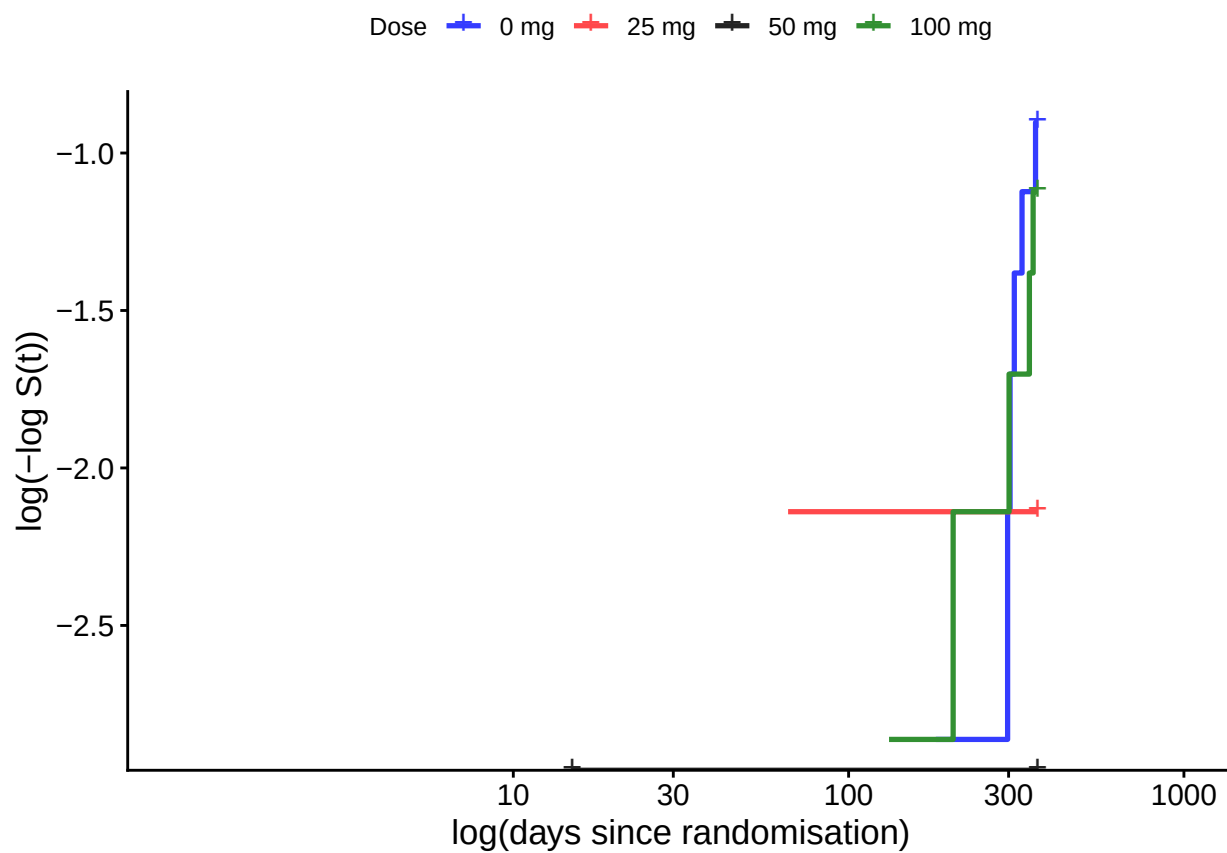
**Global Schoenfeld p-value:** 0.363 — No evidence against PH.

### 2.2.1 Scaled Schoenfeld residuals

Global Schoenfeld Test p: 0.3626



### 2.2.2 Complementary log-log survival



### 2.3 Overall survival at 24 months (OS24M)

34 events in 54 subjects.

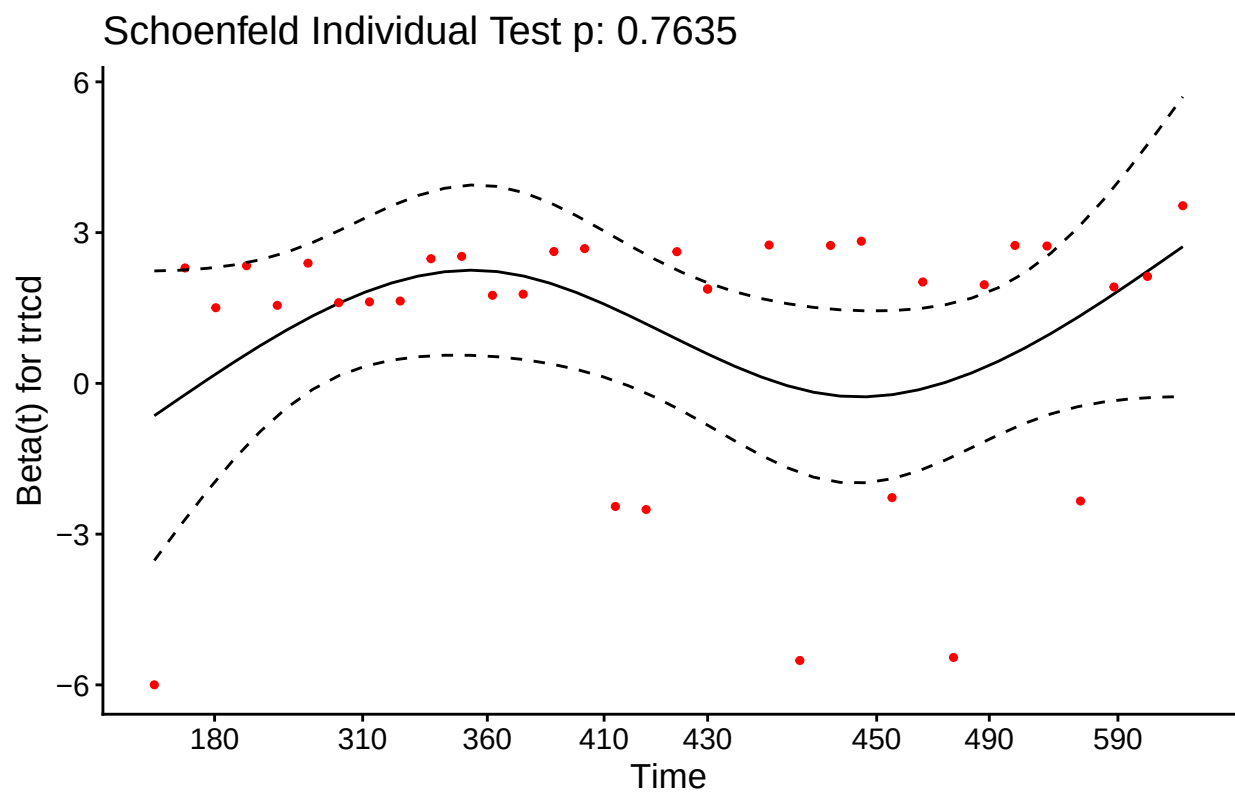
Table 3: Scaled Schoenfeld test — Overall survival at 24 months

|        | Term             | Chi-square | df | p-value |
|--------|------------------|------------|----|---------|
| trted  | trted            | 1.156      | 3  | 0.7635  |
| GLOBAL | GLOBAL (overall) | 1.156      | 3  | 0.7635  |

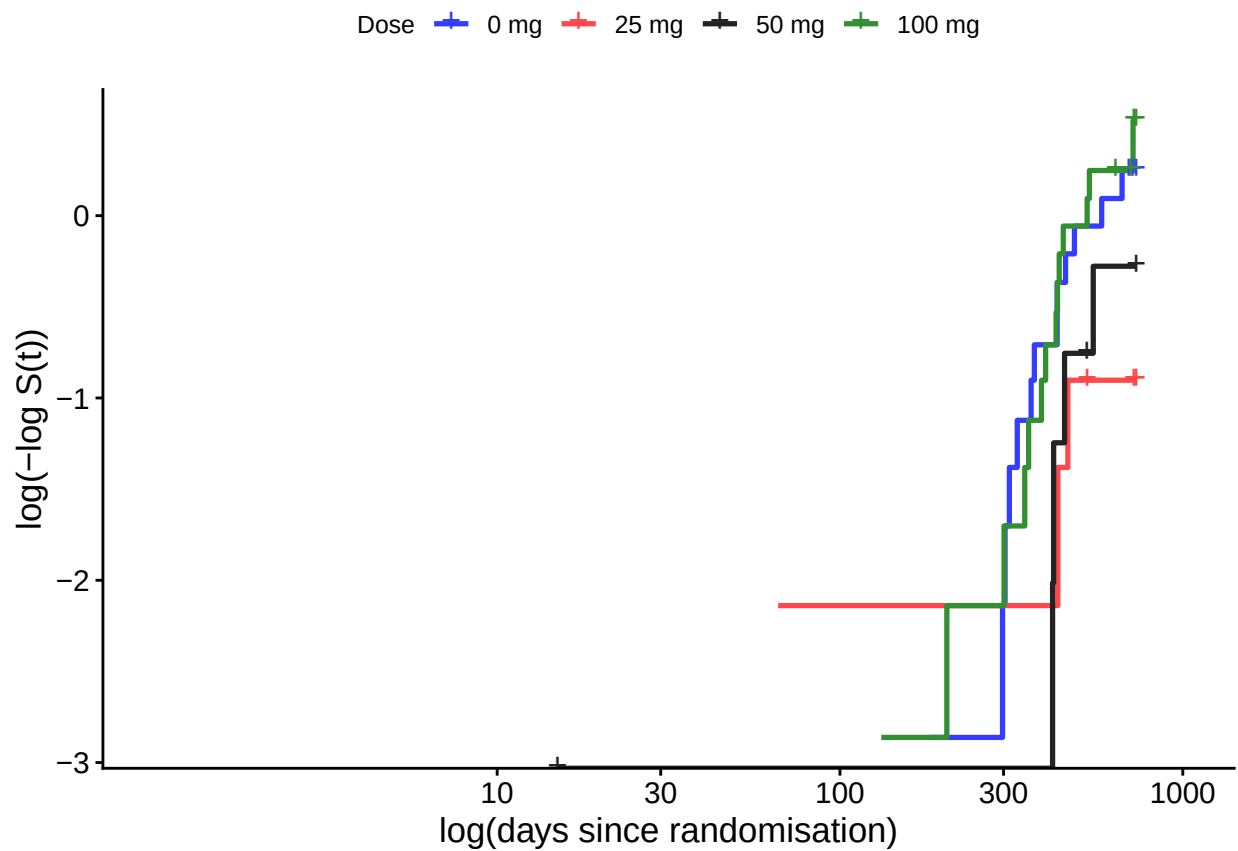
**Global Schoenfeld p-value:** 0.764 — No evidence against PH.

### 2.3.1 Scaled Schoenfeld residuals

Global Schoenfeld Test p: 0.7635



### 2.3.2 Complementary log-log survival



### 2.4 Progression-free survival (PFS)

47 events in 54 subjects.

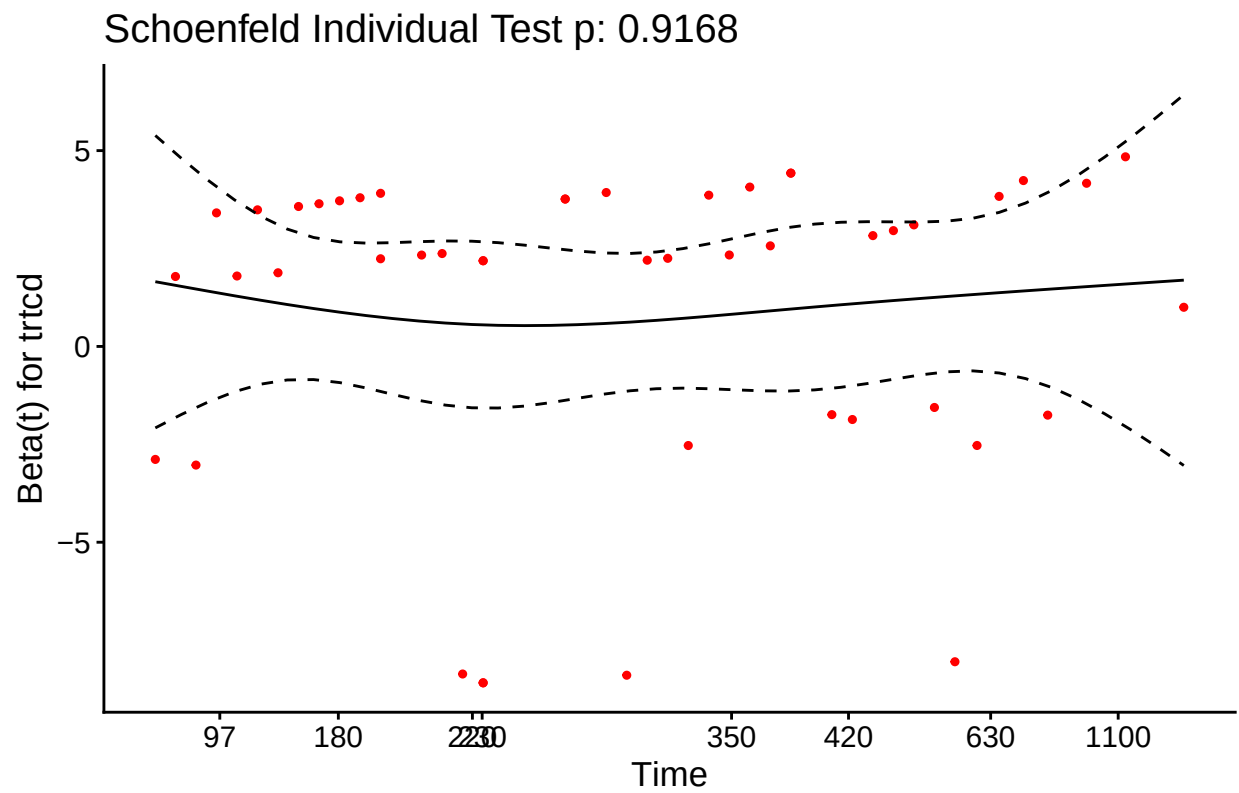
Table 4: Scaled Schoenfeld test — Progression-free survival

|        | Term             | Chi-square | df | p-value |
|--------|------------------|------------|----|---------|
| trtcd  | trtcd            | 0.509      | 3  | 0.9168  |
| GLOBAL | GLOBAL (overall) | 0.509      | 3  | 0.9168  |

**Global Schoenfeld p-value:** 0.917 — No evidence against PH.

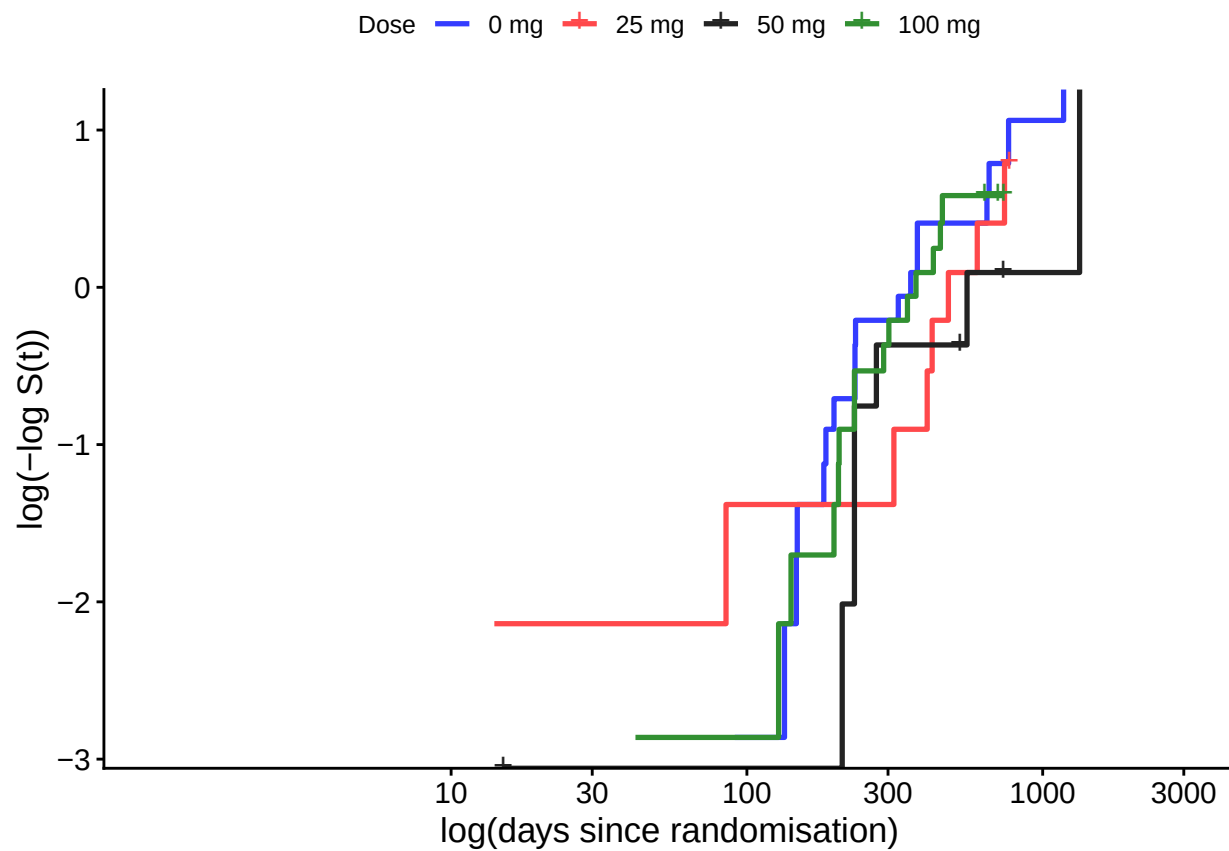
### 2.4.1 Scaled Schoenfeld residuals

Global Schoenfeld Test p: 0.9168





### 2.4.2 Complementary log-log survival



### 2.5 6-month progression-free survival (PFS6M)

9 events in 54 subjects.

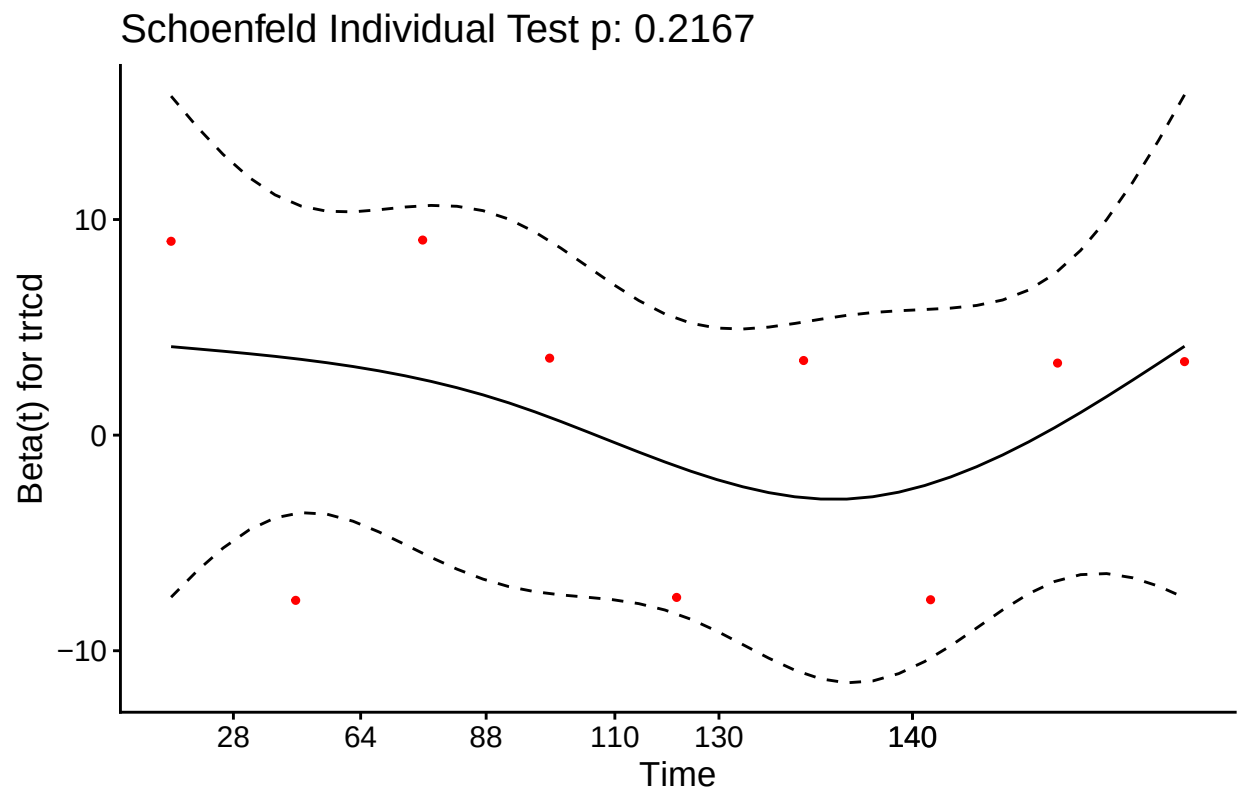
Table 5: Scaled Schoenfeld test — 6-month progression-free survival

|        | Term             | Chi-square | df | p-value |
|--------|------------------|------------|----|---------|
| trted  | trted            | 4.451      | 3  | 0.2167  |
| GLOBAL | GLOBAL (overall) | 4.451      | 3  | 0.2167  |

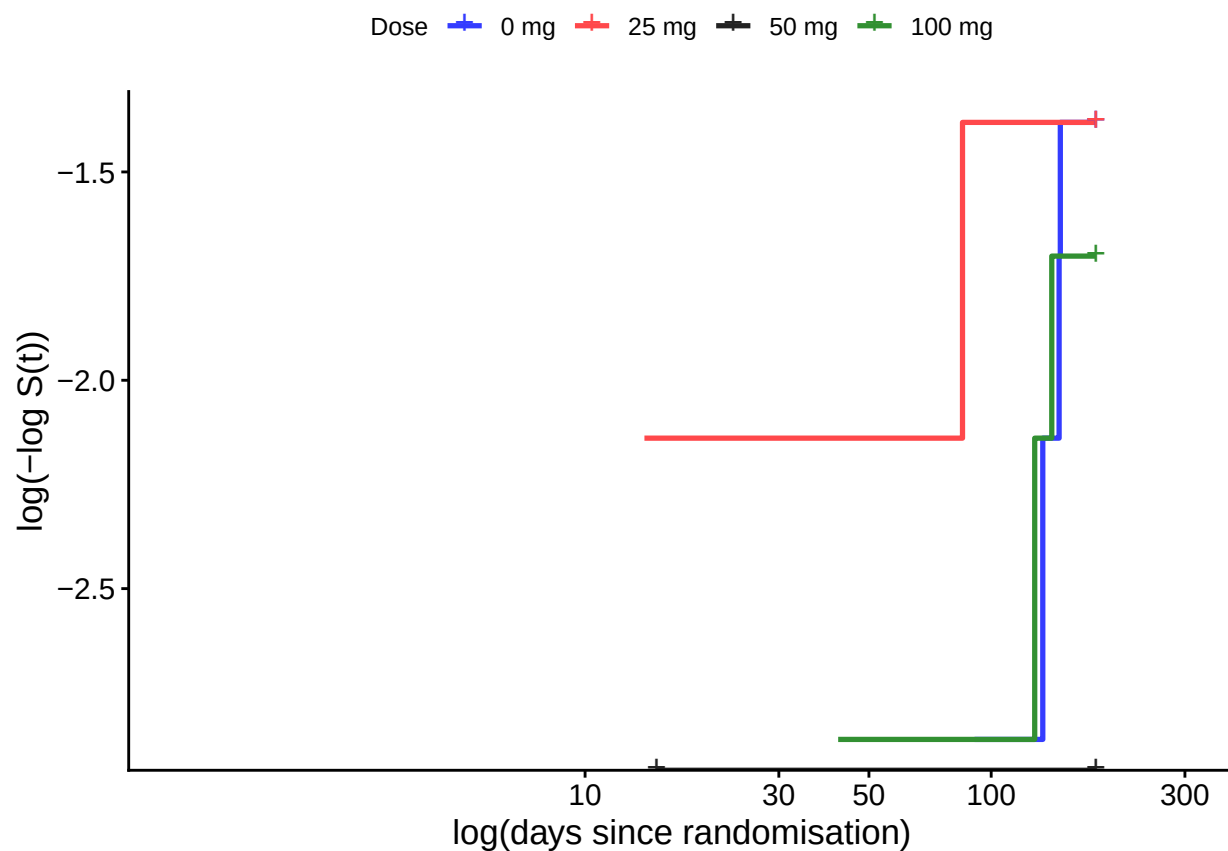
**Global Schoenfeld p-value:** 0.217 — No evidence against PH.

### 2.5.1 Scaled Schoenfeld residuals

Global Schoenfeld Test p: 0.2167



### 2.5.2 Complementary log-log survival



## 3 Summary

Table 6: Global proportional-hazards test by endpoint

| PARAMCD | Parameter                         | Global Schoenfeld p | Conclusion             |
|---------|-----------------------------------|---------------------|------------------------|
| OS      | Overall survival                  | 0.7635              | No evidence against PH |
| OS12M   | Overall survival at 12 months     | 0.3626              | No evidence against PH |
| OS24M   | Overall survival at 24 months     | 0.7635              | No evidence against PH |
| PFS     | Progression-free survival         | 0.9168              | No evidence against PH |
| PFS6M   | 6-month progression-free survival | 0.2167              | No evidence against PH |

Endpoints flagged as borderline or non-proportional should be interpreted with the corresponding plots; for those a time-averaged hazard ratio is reported in the main document with this caveat, and a time-varying-coefficient or stratified sensitivity analysis can be added at database lock.

Generated from `reports/model_fit/impress_tte_ph_check.Rmd` — sham-randomised data.